

- Description

This module manages a solar panel voltage monitoring system, ensuring that:

1. **The voltage output of a solar panel is continuously measured and compared against optimal performance levels.**
2. **If the voltage drops below the minimum threshold, a fault or shading alert is triggered.**
3. **If the voltage exceeds the maximum safe limit, a risk of electrical damage is detected.**
4. **A log of all voltage readings, alerts, and maintenance recommendations is maintained for system optimization.**

This system helps maximize solar panel efficiency, detect potential failures, and ensure safe operation.

- Specification

**module** SolarPanelVoltageMonitoring

**var** currentVoltage, maxVoltage, minVoltage: real

**var** voltageLog: sequence of string

**process** updateVoltage(changeAmount: real)

**functional scenario form:**

currentVoltage <= maxVoltage && voltageLog = voltageLog  $\oplus$  ("Voltage Updated: " + currentVoltage + " V") Voltage output updated successfully.

||

currentVoltage > maxVoltage && voltageLog = voltageLog  $\oplus$  ("Overvoltage Warning: " + currentVoltage + " V, Risk of Electrical Damage."), Overvoltage warning triggered, risk of electrical damage.

||

currentVoltage < minVoltage && voltageLog = voltageLog  $\oplus$  ("Low Voltage Alert: " + currentVoltage + " V, Possible Fault or Shading Issue."), Low voltage alert triggered, potential fault detected.

**end process**

**end module**

- Java Code

```
package code;
import java.util.*;
class SolarPanel {
    private double maxVoltage;
    private double minVoltage;
    private double currentVoltage;
    private List<String> voltageLog;
    public SolarPanel(double maxVoltage, double minVoltage, double initialVoltage) {
        this.maxVoltage = maxVoltage;
```

```

        this.minVoltage = minVoltage;
        this.currentVoltage = initialVoltage;
        this.voltageLog = new ArrayList<>();
    }
    public void updateVoltage(double changeAmount) {
        currentVoltage += changeAmount;
        String message = String.format("Current Voltage: %.2f V", currentVoltage);
        if (currentVoltage > maxVoltage) {
            message += " | WARNING: Overvoltage Detected! Risk of Electrical Damage.";
        } else if (currentVoltage < minVoltage) {
            message += " | ALERT: Low Voltage! Possible Fault or Shading Issue.";
        }
        System.out.println(message);
        voltageLog.add(message);
    }
    public void printVoltageLog() {
        System.out.println("\nVoltage Log:");
        for (String log : voltageLog) {
            System.out.println(log);
        }
    }
}

public class c31_SolarPanelSystem {
    public static void main(String[] args) {
        SolarPanel panel = new SolarPanel(50.0, 20.0, 35.0);
        // testcase-VT:
        //     panel.updateVoltage(10.0);
        //     panel.updateVoltage(-15.0);
        //     panel.updateVoltage(-12.0);
        //     panel.updateVoltage(8.0);
        //     panel.updateVoltage(87.0);
        //     panel.updateVoltage(30.0);
        //     panel.printVoltageLog();

        // testcase-FT:
        //     panel.updateVoltage(1.7976930277114552E308);
        //     panel.updateVoltage(-1.7976931348623157E308);
        //     panel.updateVoltage(0.0);
        //     panel.updateVoltage(-1.2971589679006513E308);
        //     panel.updateVoltage(4.0183728896922353E307);
        //     panel.updateVoltage(1.1596923673565998E291);
    }
}

```

```
//      panel.printVoltageLog();

      // testcase-Z3:
//      panel.updateVoltage(10.0);
//      panel.updateVoltage(15.0);
//      panel.updateVoltage(0);
//      panel.updateVoltage(8.0);
//      panel.updateVoltage(87.0);
//      panel.updateVoltage(30.0);
//      panel.printVoltageLog();

      // testcase-UVT:
panel.updateVoltage(10);
panel.updateVoltage(15);
panel.updateVoltage(12);
panel.updateVoltage(8);
panel.updateVoltage(87);
panel.updateVoltage(30);
//      panel.printVoltageLog();
    }
}
```

- Coverage report

1. Test case number: 6
2. V-Method testing branch coverage: 100%

| Element                     | Coverage                       | Covered Branches | Missed Branches | Total Branches |
|-----------------------------|--------------------------------|------------------|-----------------|----------------|
| ▼ VibrationTestingTool      | <div><div></div></div> 0.7 %   | 6                | 838             | 844            |
| ▼ src                       | <div><div></div></div> 0.7 %   | 6                | 838             | 844            |
| ▼ code                      | <div><div></div></div> 0.7 %   | 6                | 838             | 844            |
| ▼ c31_SolarPanelSystem.java | <div><div></div></div> 100.0 % | 6                | 0               | 6              |
| > c31_SolarPanelSystem      |                                | 0                | 0               | 0              |
| > SolarPanel                | <div><div></div></div> 100.0 % | 6                | 0               | 6              |

3. Vibration testing instruction coverage: 97.3%

| Element                     | Coverage                       | Covered Instructions | Missed Instructions | Total Instructions |
|-----------------------------|--------------------------------|----------------------|---------------------|--------------------|
| ▼ VibrationTestingTool      | <div><div></div></div> 0.5 %   | 109                  | 19,973              | 20,082             |
| ▼ src                       | <div><div></div></div> 0.5 %   | 109                  | 19,973              | 20,082             |
| ▼ code                      | <div><div></div></div> 0.5 %   | 109                  | 19,973              | 20,082             |
| ▼ c31_SolarPanelSystem.java | <div><div></div></div> 97.3 %  | 109                  | 3                   | 112                |
| > c31_SolarPanelSystem      | <div><div></div></div> 90.3 %  | 28                   | 3                   | 31                 |
| > SolarPanel                | <div><div></div></div> 100.0 % | 81                   | 0                   | 81                 |

4. Fuzz testing branch coverage: 83.3%

| Element                     | Coverage                      | Covered Branches | Missed Branches | Total Branches |
|-----------------------------|-------------------------------|------------------|-----------------|----------------|
| ▼ VibrationTestingTool      | <div><div></div></div> 0.6 %  | 5                | 839             | 844            |
| ▼ src                       | <div><div></div></div> 0.6 %  | 5                | 839             | 844            |
| ▼ code                      | <div><div></div></div> 0.6 %  | 5                | 839             | 844            |
| ▼ c31_SolarPanelSystem.java | <div><div></div></div> 83.3 % | 5                | 1               | 6              |
| > c31_SolarPanelSystem      |                               | 0                | 0               | 0              |
| > SolarPanel                | <div><div></div></div> 83.3 % | 5                | 1               | 6              |

5. Fuzz testing instruction coverage: 97.3%

| Element                     | Coverage | Covered Instructions | Missed Instructions | Total Instructions |
|-----------------------------|----------|----------------------|---------------------|--------------------|
| ✓ VibrationTestingTool      | 0.5 %    | 109                  | 19,973              | 20,082             |
| ✓ src                       | 0.5 %    | 109                  | 19,973              | 20,082             |
| ✓ code                      | 0.5 %    | 109                  | 19,973              | 20,082             |
| ✓ c31_SolarPanelSystem.java | 97.3 %   | 109                  | 3                   | 112                |
| > c31_SolarPanelSystem      | 90.3 %   | 28                   | 3                   | 31                 |
| > SolarPanel                | 100.0 %  | 81                   | 0                   | 81                 |

6. Z3 branch coverage: 83.3%

| Element                     | Coverage | Covered Branches | Missed Branches | Total Branches |
|-----------------------------|----------|------------------|-----------------|----------------|
| ✓ VibrationTestingTool      | 0.6 %    | 5                | 835             | 840            |
| ✓ src                       | 0.6 %    | 5                | 835             | 840            |
| ✓ code                      | 0.6 %    | 5                | 835             | 840            |
| ✓ c31_SolarPanelSystem.java | 83.3 %   | 5                | 1               | 6              |
| > SolarPanel                | 83.3 %   | 5                | 1               | 6              |
| > c31_SolarPanelSystem      |          | 0                | 0               | 0              |

7. Z3 instruction coverage: 94.6%

| Element                     | Coverage | Covered Instructions | Missed Instructions | Total Instructions |
|-----------------------------|----------|----------------------|---------------------|--------------------|
| ✓ VibrationTestingTool      | 0.5 %    | 106                  | 22,292              | 22,398             |
| ✓ src                       | 0.5 %    | 106                  | 22,292              | 22,398             |
| ✓ code                      | 0.5 %    | 106                  | 22,292              | 22,398             |
| ✓ c31_SolarPanelSystem.java | 94.6 %   | 106                  | 6                   | 112                |
| > c31_SolarPanelSystem      | 90.3 %   | 28                   | 3                   | 31                 |
| > SolarPanel                | 96.3 %   | 78                   | 3                   | 81                 |

8. Unimproved V-method branch coverage: 50%

| Element                     | Coverage | Covered Branches | Missed Branches | Total Branches |
|-----------------------------|----------|------------------|-----------------|----------------|
| ✓ VibrationTestingTool      | 0.4 %    | 3                | 837             | 840            |
| ✓ src                       | 0.4 %    | 3                | 837             | 840            |
| ✓ code                      | 0.4 %    | 3                | 837             | 840            |
| ✓ c31_SolarPanelSystem.java | 50.0 %   | 3                | 3               | 6              |
| > SolarPanel                | 50.0 %   | 3                | 3               | 6              |
| > c31_SolarPanelSystem      |          | 0                | 0               | 0              |

9. Unimproved V-method instruction coverage: 77.3%

| Element                     | Coverage | Covered Instructions | Missed Instructions | Total Instructions |
|-----------------------------|----------|----------------------|---------------------|--------------------|
| ✓ VibrationTestingTool      | 0.4 %    | 85                   | 22,311              | 22,396             |
| ✓ src                       | 0.4 %    | 85                   | 22,311              | 22,396             |
| ✓ code                      | 0.4 %    | 85                   | 22,311              | 22,396             |
| ✓ c31_SolarPanelSystem.java | 77.3 %   | 85                   | 25                  | 110                |
| > SolarPanel                | 72.8 %   | 59                   | 22                  | 81                 |
| > c31_SolarPanelSystem      | 89.7 %   | 26                   | 3                   | 29                 |